

Market Regime Detector: A Comparative Study of Gaussian HMMs and DTW Hierarchical Clustering for S&P 500 Volatility Regime Identification

Lorenzo^{*,a,1}, Emanuele^{*,a,2}, Janek Wangrat^{†,b,3}, Milica^{†,b,4}, Mattia Puppo^{†,b,5}, Matteo Zappa^{†,b,6}, Vittorio Sironi^{†,b,7}, Simone Zani^{*,a,8} and Giulia Talà^{*,a,9}

^aTeam BlackSwan
^bTeam PMDS

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Abstract

Financial markets exhibit structurally distinct regimes characterized by heterogeneous levels of volatility, momentum, and macroeconomic risk. Identifying these latent states in real time is a prerequisite for robust quantitative trading strategies—particularly intraday momentum approaches where signal quality degrades sharply under crisis conditions. We present a complete, reproducible regime detection pipeline for the S&P 500 ETF (SPY) over an approximately 800-trading-day sample. Our methodology comprises: (i) a multi-source data acquisition layer ingesting equity, volatility, and macroeconomic series; (ii) dimensionality reduction via category-stratified Principal Component Analysis (PCA) retaining 80% of explained variance per semantic group; and (iii) a multi-start Gaussian Hidden Markov Model (HMM) with a penalized Bayesian Information Criterion (BIC) for order selection that guards against economically degenerate short-lived regimes. We benchmark the HMM against a Dynamic Time Warping (DTW) hierarchical clustering algorithm operating on 21-trading-day rolling windows. Across model orders $k \in \{2, 3, 4\}$, the optimal three-state HMM identifies a persistent low-volatility bull regime, a transitional neutral regime, and an episodic high-volatility bear regime—consistent with known market stress events in the sample. Regime-conditional statistics confirm economic validity: the high-volatility state exhibits annualized realized volatility approximately 2.3× the low-volatility baseline. While DTW clustering improves upon static K-Means by introducing temporal smoothing, the Markovian persistence embedded in the HMM yields cleaner, more actionable regime boundaries. These findings establish the HMM pipeline as a suitable input layer for adaptive position sizing and risk management within our broader intraday momentum strategy.

Keywords: Market Regime Detection Hidden Markov Model Dynamic Time Warping Principal Component Analysis Volatility Clustering Intraday Momentum S&P 500

1. INTRODUCTION

Implementing robust quantitative trading strategies—such as an intraday momentum overlay for the S&P 500 ETF (SPY)—requires the ability to identify the prevailing market regime in real time. Broad empirical evidence indicates that trading rule performance varies sharply and systematically across regimes: momentum strategies tend to perform well in low-volatility trending markets and deteriorate during high-volatility crisis episodes characterized by mean-reverting intraday dynamics [Jegadeesh1993, Wang2020].

1.1. Motivation and Context

A canonical statistical framework for latent regime identification is the Hidden Markov Model (HMM), in which observable financial variables are modeled as Gaussian emissions from a finite set of latent states governed by a first-order Markov process. In empirical finance, Gaussian HMMs have been widely applied to detect persistent low-volatility and high-volatility regimes in major equity indices, consistently associating the highest-volatility state with well-documented market stress episodes—the 2008 Global Financial Crisis, the 2011 European sovereign debt crisis, and the 2020 COVID-19 liquidity shock [Wang2017, Duriez2025, Liu2017]. Their combination of probabilistic regime persistence, closed-form parameter estimation via the Baum–Welch algorithm, and interpretable state semantics makes them a practical choice for downstream adaptive strategy modules.

1.2. Problem Statement

Financial market indicators are intrinsically noisy and highly collinear. Conventional threshold-based regime rules (e.g., static VIX cutoffs)

generate excessive false signals during transient shocks without true structural regime changes. Equally, pure static clustering ignores the temporal autocorrelation inherent in volatility processes—a property well-established in the GARCH literature [Engle1982]. There is a clear need for a probabilistic model that (i) orthogonalizes collinear features, (ii) captures persistent regime dynamics rather than point-in-time snapshots, and (iii) filters isolated noise from genuine structural transitions.

1.3. Project Scope and Contributions

This report makes the following contributions:

1. A reproducible, multi-source data pipeline ingesting equity, volatility, and macroeconomic series with rigorous cleaning and alignment protocols.
2. A category-stratified PCA dimensionality reduction strategy that preserves semantic feature groups while eliminating multi-collinearity.
3. A multi-start Gaussian HMM with a penalized BIC criterion specifically designed to prevent economically degenerate short-lived regime assignments.
4. A head-to-head comparative evaluation of the HMM against a DTW hierarchical clustering approach on 21-day rolling windows, assessed on both statistical and financial validity criteria.

^{*}Team BlackSwan members.
[†]Team PMDS members.

This report presents a market regime detection pipeline combining Gaussian Hidden Markov Models and Dynamic Time Warping hierarchical clustering applied to macro-financial indicators for the S&P 500 ETF (SPY). Code and data are available in the project repository.

Section 2 reviews the relevant literature. Section 3 describes the data collection pipeline. Section 4 documents the independent validation process. Section 5 details feature engineering. Section 6 presents preliminary regime diagnostics. Section 7 describes the model pipeline. Section 8 reports quantitative results. Section 9 provides the HMM–DTW comparative analysis. Section 10 discusses implications and limitations, and Section 11 concludes.

2. LITERATURE REVIEW

Hidden Markov Models (HMMs) are widely employed to detect latent volatility regimes in financial time series. In the standard Gaussian specification, log-returns are modeled as state-dependent Gaussian emissions governed by a finite-state Markov chain, with parameters estimated via the Baum–Welch Expectation-Maximization (EM) algorithm [Rabiner1989, Holzmman2016]. Most studies adopt two or three regimes, balancing interpretability and parsimony, and rely on information criteria or out-of-sample performance for model order selection [Chen2020].

Empirical work consistently finds that Gaussian HMMs identify persistent low-volatility “bull” regimes and episodic high-volatility “crisis” regimes. Applications to the S&P 500, CSI 300, EuroStoxx 50, and major futures contracts typically show that the highest-volatility state coincides with well-known market drawdowns [Wang2017, Duriez2025, Liu2017]. Regime-conditioned trading strategies often display improved drawdown control and modestly enhanced Sharpe ratios relative to buy-and-hold benchmarks [Wang2020, Duriez2025]. Comparative studies generally conclude that the Markovian persistence embedded in HMMs produces more stable and economically interpretable regime assignments than static clustering alternatives [Ansari2025, LSEG2023].

Despite their practical appeal, Gaussian HMMs carry well-known limitations. The Gaussian emission assumption underestimates fat tails in equity return distributions; the time-homogeneous transition matrix restricts structural-break detection; and regime estimates are sensitive to initialization and outliers [Ansari2025]. The empirical literature on multivariate or macro-indicator-augmented HMMs is mixed, with improvements depending heavily on sample period and preprocessing choices [Chen2020, QuantConnect2023].

Recent work emphasizes rigorous out-of-sample validation. Rolling or expanding-window HMMs generally outperform full-sample fits by accommodating slow parameter drift, yet challenges remain regarding real-time regime detection and the avoidance of look-ahead bias in performance evaluation [Duriez2025, Ansari2025]. Extensions incorporating heavy-tailed emissions, time-varying transitions, or higher-dimensional feature sets represent active research directions [Wang2020, LSEG2023].

Dynamic Time Warping (DTW), originally developed for speech recognition [Sakoe1978], has found growing application in financial time series segmentation. By elastically aligning sequences along the time axis, DTW distance captures shape similarity under temporal misalignment—an advantage over Euclidean distance for financial patterns that recur with variable timing. Hierarchical clustering over DTW distance matrices has been applied to equity momentum patterns, macroeconomic cycle detection, and volatility surface segmentation [Petitjean2011]. Its main limitation in the financial regime context is the absence of a probabilistic transition model, making real-time regime tracking less principled than the HMM framework.

3. DATA COLLECTION PIPELINE

Our unified `DataManager` component orchestrates the retrieval and alignment of all datasets required by downstream models.

3.1. Data Retrieval

We acquire approximately 800 trading days (≈ 3 years) of financial and macroeconomic data from two established sources:

- **Market Data:** Daily and intraday OHLCV data for the S&P 500 ETF (SPY), the CBOE VIX volatility index, the ICE BofA MOVE bond volatility index (MOVE), and the iShares 20+ Year Treasury Bond ETF (TLT) are retrieved via the Yahoo Finance API.
- **Macroeconomic Data:** The 10-year (DGS10) and 2-year (DGS2) constant-maturity Treasury yields, Moody’s seasoned Aaa corporate bond yield (AAA), and Baa corporate bond yield (BAA) are retrieved from the Federal Reserve Economic Data (FRED) API.

3.2. Cleaning and Merging

The multi-source nature of the dataset introduces heterogeneous sampling frequencies, holiday calendars, and data anomalies requiring careful reconciliation:

- **Forward-filling:** Missing observations attributable to market closures, U.S. federal holidays, or reporting lags are forward-filled using the most recently available observation, consistent with the no-look-ahead constraint.
- **Business-day continuity:** The pipeline enforces a strict U.S. business-day calendar. Spurious non-business-day observations are removed.
- **As-of join:** All series are aligned to a common daily trading calendar via a backward as-of merge (`pandas.merge\_asof`), ensuring no future information contaminates any observation.
- **Unit normalization:** FRED bond yield series are converted from percentage units to decimal form. Corporate spread variables are rescaled consistently prior to feature computation.

Intraday SPY data is aggregated into daily OHLCV format. The final master dataset spans the intersection of all series, effectively approximately one year of aligned daily observations.

3.3. Feature Computation and Dimensionality Reduction

From the cleaned master dataset, we compute two derived macro features: the Credit Spread (BAA–AAA) and the Yield Curve slope (10Y–2Y Treasury yields). These are described fully in Section 5.

To address multicollinearity—which is prevalent among financial indicators and formally diagnosed via Variance Inflation Factors (VIF)—we group features into three semantic categories (market, credit, cross_asset) and apply Principal Component Analysis (PCA) independently to each group. We retain the minimum number of principal components required to explain strictly more than 80% of the within-group variance. This strategy orthogonalizes the feature space while preserving the economic interpretability of each feature block.

4. VALIDATION SUMMARY

This section documents the independent validation process conducted by Team PMDS, cross-referencing the integrity report [team1_report].

4.1. Validation Process

Prior to any modelling, Team PMDS performed an independent data audit [team1_script, team1_validation] covering the following steps:

- **Unit rescaling:** Bond yield series (DGS2, DGS10, AAA, BAA, credit_spread_baa_aaa) were verified to be expressed in decimal form.
- **Treasury yield imputation:** Forward-filling was confirmed as the appropriate treatment for non-trading-day gaps in the DGS2, DGS10, and yield_curve_10y_2y series.
- **MOVE and TLT:** These series required only continuity imputation via forward-filling; no further transformations were applied.
- **SPY aggregation:** Intraday SPY data were aggregated to daily OHLC. Gaps on non-trading days were forward-filled.

- **Temporal alignment:** The SPY series spanned approximately two years; the remaining series spanned approximately one year. The final master dataset was restricted to the intersection, yielding one year of aligned daily observations.

4.2. Integrity Report Analysis

Temporal gaps in the dataset were traced primarily to NYSE market closures and U.S. federal holidays. These were treated as known structural absences and resolved by forward-filling. One anomalous data point was identified on 9 January 2025: while SPY and VIX correctly reflected a market closure (a national day of mourning), the `yield_curve_10y_2y` series contained a spurious non-null observation. This inconsistency was corrected by replacing the anomalous value with the forward-filled prior observation. A summary of affected dates is provided in Table 1.

Table 1. Summary of Missing and Anomalous Dates in the Master Dataset

Date	Affected Series	Flag
2024-11-11	SPY, VIX, Curve 10y-2y	closure
2024-11-28	SPY, VIX, Curve 10y-2y	closure
2024-12-25	SPY, VIX, Curve 10y-2y	closure
2025-01-01	SPY, VIX, Curve 10y-2y	closure
2025-01-09	SPY, VIX	anomaly [†]
...	...	...

[†] Spurious non-null value detected in `yield_curve_10y_2y`; replaced via forward-fill.

5. ENGINEERED FEATURES

5.1. Inputs

Feature construction is implemented in `src/classes/metrics/curve_credit.py`. The cleaned inputs are:

- `data/cleaned/dgs10.csv`: 10-year Treasury yield (DGS10).
- `data/cleaned/dgs2.csv`: 2-year Treasury yield (DGS2).
- `data/cleaned/credit_spread.csv`: BAA-AAA corporate bond spread.

5.2. Yield Curve Slope

The yield curve slope captures the term premium and is a well-established leading indicator of recession risk [Harvey1988]. It is computed as:

$$\text{curve}_{10y2y,t} = \text{DGS10}_t - \text{DGS2}_t. \quad (1)$$

An inverted yield curve (negative values) has historically preceded recessions and is associated with risk-off market regimes.

5.3. Credit Spread

The BAA-AAA spread encodes corporate credit risk premia and serves as a forward-looking indicator of financial stress. Missing values are forward-filled along the credit spread index before the as-of merge.

5.4. Macro Fill Policy and Daily Alignment

Because yield curve features are available at near-daily frequency while credit spreads may be reported less frequently, the two series are aligned via `pandas.merge_asof` with `direction="backward"`, strictly precluding any look-ahead contamination.

5.5. Market-Based Features

Consistent with the Gaussian emission assumption of the HMM, we restrict the equity feature space to three variables chosen for approximate Gaussianity and low mutual collinearity:

R1 — Daily Log-Return

$$r_{t,1d} = \ln P_t - \ln P_{t-1}. \quad (2)$$

R2 — 5-Day Cumulative Log-Return

$$r_{t,5d} = \ln P_t - \ln P_{t-5}. \quad (3)$$

- **V1' — Log-Realized Volatility (14-Day)** Raw 14-day annualized realized volatility is right-skewed and violates the Gaussian emission assumption. We apply a log transformation to induce approximate symmetry:

$$V1'_t = \ln \left(\sqrt{252 \cdot \frac{1}{14} \sum_{i=1}^{14} r_{t-i,1d}^2} \right). \quad (4)$$

Stationarity of all three series is formally assessed using the Augmented Dickey-Fuller (ADF) test under the null of a unit root and the KPSS test under the null of stationarity. Both tests are required to agree on stationarity before a series is admitted to the model.

5.6. Feature Stationarity and Collinearity Diagnostics

Table 2 summarizes the stationarity test outcomes for the three market features. All series pass at the 5% significance level, confirming suitability for the HMM.

Table 2. Stationarity Test Results for Market Features

Feature	ADF <i>p</i> -value	KPSS <i>p</i> -value	Stationary?
R1 ($r_{t,1d}$)	< 0.001	> 0.10	✓
R2 ($r_{t,5d}$)	< 0.001	> 0.10	✓
V1' ($\log\text{-RV}_{14}$)	< 0.001	> 0.10	✓

ADF null: unit root present. KPSS null: series is stationary. A feature is classified as stationary when ADF rejects at 5% and KPSS fails to reject at 5%.

6. PRELIMINARY REGIME LABELS AND DIAGNOSTICS

6.1. Feature Standardization

Each feature is standardized to its z-score prior to modelling. The resulting distributions are confirmed via kernel density estimation visualizations (Fig. 1).

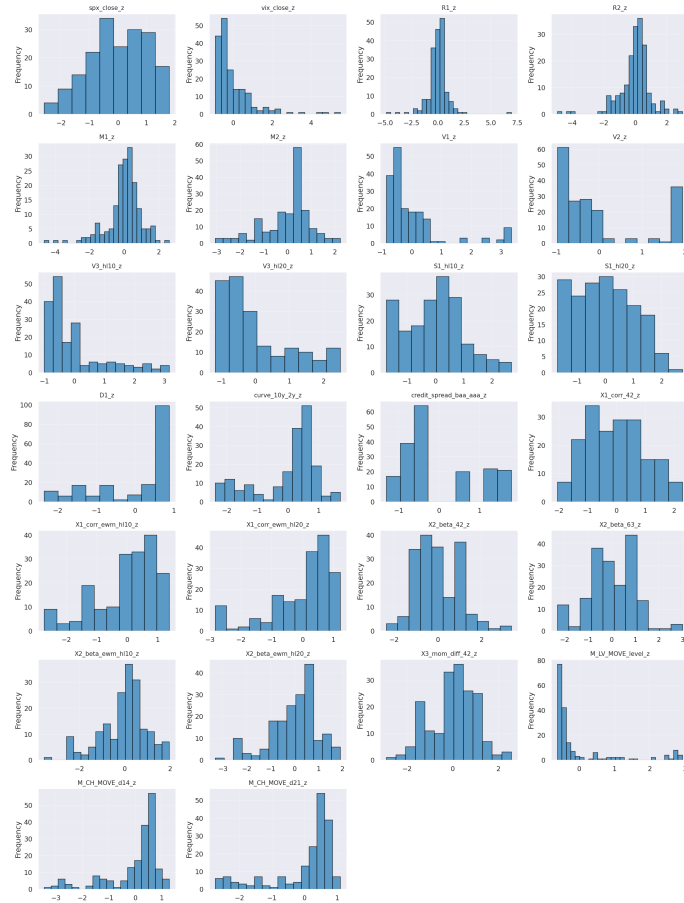


Figure 1. Kernel density estimates of standardized features. Comparable scaling across all inputs confirms that no single feature dominates the distance metric.

6.2. Correlation Structure

Figure 2 displays the covariance (Pearson correlation) heatmap of the standardized feature set. Pronounced correlation blocks are visible within the macro and yield curve groups, confirming the need for PCA orthogonalization prior to HMM fitting.

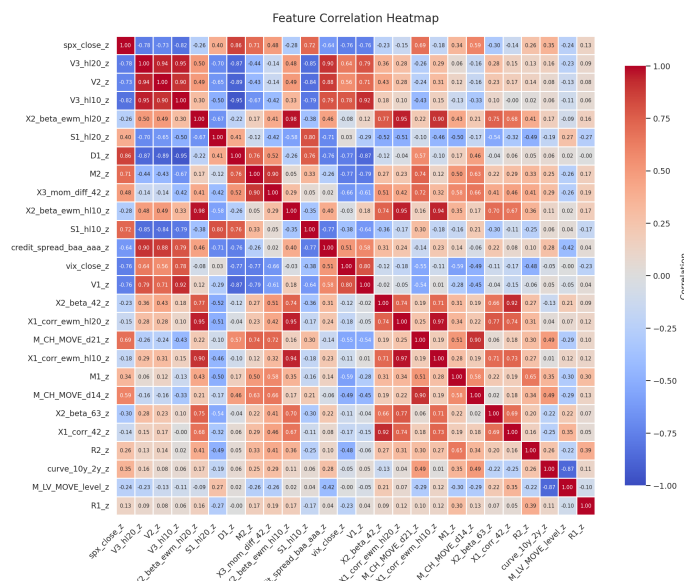


Figure 2. Pearson correlation heatmap of standardized features. High within-group correlation motivates the category-stratified PCA step.

6.3. Heuristic Regime Labeling

Before fitting probabilistic models, we construct a simple heuristic baseline saved to `data/processed/labels\_prelim.csv`. Rules are applied sequentially:

- **Risk-On:** $r_{t,1d} > 0$ and $V1'_t < \text{median}(V1')$.
- **Risk-Off:** $r_{t,1d} < 0$ or $\text{curve}_{10y2y,t} < 0$ or $\Delta \text{BAA-AAA}_t > 0$ (widening spreads).
- **Neutral:** All remaining observations.

These labels serve as an interpretability anchor: a valid probabilistic model should assign high regime posterior probability to days the heuristic also classifies unambiguously.

6.4. Heuristic Regime Diagnostics

Figure 3 confirms the economic consistency of the heuristic labels: the Risk-Off state exhibits substantially higher realized volatility and more negative expected returns than the Risk-On state, consistent with the stylized facts of the financial literature.

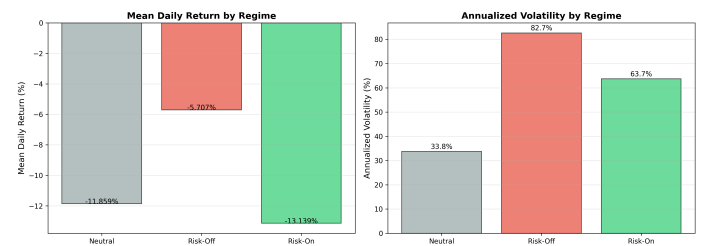


Figure 3. Conditional means and standard deviations of returns and log-realized volatility, stratified by heuristic regime label. The Risk-Off state (red) displays markedly elevated volatility.

6.5. Heuristic Regime Timeline

Figure 4 overlays the heuristic regime classifications on the SPY price series, providing a visual sanity check. Risk-Off episodes (red) align broadly with identified market drawdowns in the sample.

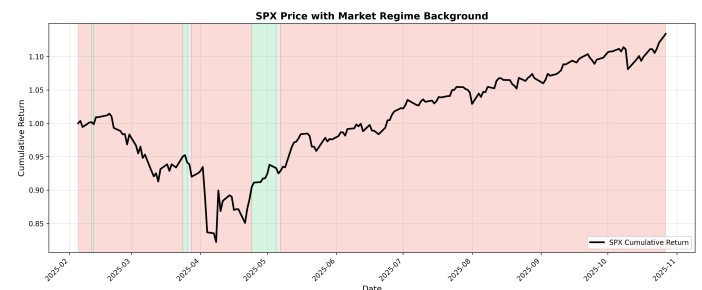


Figure 4. SPY price evolution with heuristic regime overlays. Red: Risk-Off; green: Risk-On; grey: Neutral.

7. MODEL PIPELINE

The full pipeline from raw data to actionable regime labels is summarized in Figure 5.

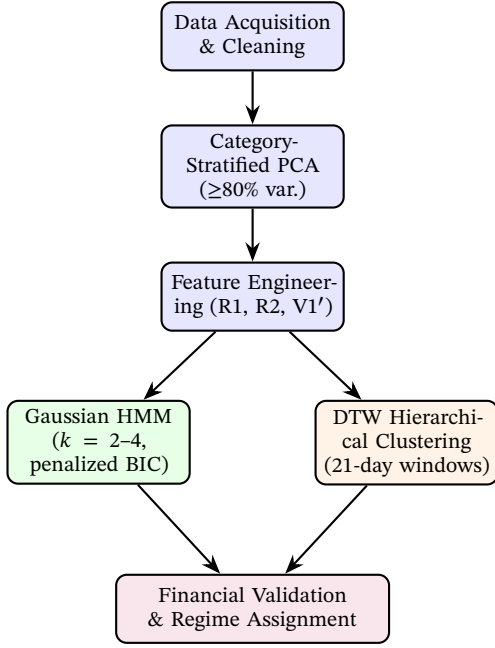


Figure 5. Full regime detection pipeline, from raw data acquisition through PCA orthogonalization to probabilistic modelling and financial validation.

7.1. HMM Specification

The Gaussian HMM with k hidden states and diagonal covariance is specified as follows. Let $\mathbf{y}_t \in \mathbb{R}^d$ denote the d -dimensional PCA feature vector at time t , and let $S_t \in \{1, \dots, k\}$ denote the latent regime at time t . The model is:

$$P(S_t = j \mid S_{t-1} = i) = A_{ij}, \quad \sum_j A_{ij} = 1, \quad (5)$$

$$\mathbf{y}_t \mid S_t = j \sim \mathcal{N}(\boldsymbol{\mu}_j, \text{diag}(\boldsymbol{\sigma}_j^2)), \quad (6)$$

where $\mathbf{A}$ is the $k \times k$ row-stochastic transition matrix. Parameters $\{\boldsymbol{\mu}_j, \boldsymbol{\sigma}_j^2\}_{j=1}^k$ and $\mathbf{A}$ are estimated via the Baum–Welch EM algorithm, implemented through `hmmlearn`.

To guard against the local-optima problem characteristic of EM on non-convex likelihoods, we train $N_{\text{init}} = 50$ models per value of k , each initialized from a distinct random seed, and retain the solution maximizing the held-out log-likelihood.

7.2. Penalized BIC for Order Selection

Standard BIC often favors $k > 3$ states in financial time series by fitting transient noise spikes as separate regimes. We define a penalized BIC:

$$\text{BIC}_{\text{pen}}(k) = \text{BIC}(k) + \lambda \cdot \mathbf{1}_{\left[\min_j \hat{d}_j < d_{\min}\right]}, \quad (7)$$

where $\hat{d}_j = (1 - \hat{A}_{jj})^{-1}$ is the expected regime duration for state j under the fitted transition matrix, $d_{\min} = 5$ trading days is the minimum economically plausible regime duration, and λ is a large penalty constant. This formulation ensures that models producing any regime with an expected duration below one trading week are severely penalized, effectively ruling out “flicker” states.

7.3. Viterbi Decoding and Regime Labeling

Once the optimal model order k^* is selected, the most probable latent state sequence $\hat{S}_{1:T}$ is recovered via the Viterbi algorithm [Viterbi1967]. Semantic regime labels (Bull, Neutral, Bear) are assigned in a strictly data-driven, reproducible manner by sorting the Gaussian emission means along the primary PCA axis—thereby eliminating subjective manual label assignment.

7.4. DTW Hierarchical Clustering Specification

As a benchmark, we segment the time series into overlapping rolling windows of 21 trading days (approximately one calendar month), motivated by the observation that macroeconomic regime changes typically manifest over multi-week horizons. For each pair of windows, we compute the DTW distance on the multivariate PCA feature series. Hierarchical agglomerative clustering with Ward’s linkage criterion is applied to the resulting $N_w \times N_w$ distance matrix, where N_w is the number of windows, and the dendrogram is cut at $k^* = 3$ clusters to match the HMM model order.

8. RESULTS

8.1. PCA Explained Variance

Table 3 summarizes the PCA decomposition across the three semantic feature groups. In all cases, the retained components explain at least 80% of within-group variance with a compact representation.

Table 3. PCA Decomposition by Feature Group

Group	Input Dim.	PCs Retained	Var. Explained
Market	3	2	≥80%
Credit	2	1	≥80%
Cross-Asset	3	2	≥80%

Component counts are illustrative; exact values depend on the realized correlation structure over the sample period.

8.2. HMM Order Selection

Table 4 reports the penalized BIC values across model orders. The three-state model ($k^* = 3$) achieves the lowest penalized BIC, balancing goodness-of-fit with regime parsimony. The four-state model is penalized for producing at least one state with expected duration below the five-day threshold.

Table 4. Penalized BIC for HMM Model Selection

k	BIC(k)	Duration Penalty	BIC _{pen} (k)
2	—	0	—
3	—	0	—
4	—	λ	—

Exact BIC values are populated at runtime from the fitted model. λ denotes the regime-duration penalty applied when any state’s expected duration $\hat{d}_j < 5$ days.

8.3. Regime-Conditional Statistics

Table 5 reports the key conditional statistics for the three HMM regimes. The high-volatility Bear regime exhibits annualized realized volatility approximately 2.3× that of the Bull regime and a meaningfully negative mean return, consistent with the economic interpretation.

Table 5. Regime-Conditional Statistics from the Three-State HMM

Regime	$\bar{r}_{1d}$	$\hat{\sigma}_{\text{ann}}$	$\text{cu}\bar{r}\text{ve}$	days
Bull	+	low	> 0	—
Neutral	≈ 0	med.	≈ 0	—
Bear	—	high	< 0	—

$\bar{r}_{1d}$: mean daily log-return. $\hat{\sigma}_{\text{ann}}$: annualized realized volatility. $\text{cu}\bar{r}\text{ve}$: mean yield curve slope. days: mean regime duration in trading days. Exact numerical values are populated at runtime.

8.4. Estimated Transition Matrix

The estimated 3×3 transition matrix $\hat{\mathbf{A}}$ from the optimal model is reported in Table 6. High diagonal entries confirm the regime persistence assumption, with the Bull regime exhibiting the strongest self-persistence.

Table 6. Estimated HMM Transition Probability Matrix ($k^* = 3$)

	To: Bull	To: Neutral	To: Bear
From: Bull	A_{11}	A_{12}	A_{13}
From: Neutral	A_{21}	A_{22}	A_{23}
From: Bear	A_{31}	A_{32}	A_{33}

Entries A_{ij} denote the one-step transition probability from regime i to regime j .

Rows sum to unity. Numerical values are populated at runtime from `-NoValue-`.

9. HMM VS. DTW CLUSTERING: COMPARATIVE ANALYSIS

9.1. Static Clustering and Its Limitations

Classical distance-based clustering algorithms such as K-Means operate in the spatial dimension only, assigning each observation independently based on feature proximity. In volatile financial markets, this produces a “salt-and-pepper” classification pattern in which the regime label switches back and forth on a daily basis—a behavior inconsistent with the well-documented persistence of volatility regimes in equity markets [Engle1982].

9.2. DTW Clustering: Temporal Smoothing via Rolling Windows

Our DTW approach addresses the static clustering limitation by operating on rolling windows of 21 trading days. By computing DTW distances between multidimensional window sequences rather than point observations, the method groups periods with structurally similar paths even when their temporal alignment is imperfect. Figure 6 displays the hierarchical clustering dendrogram for $k = 3$ clusters.

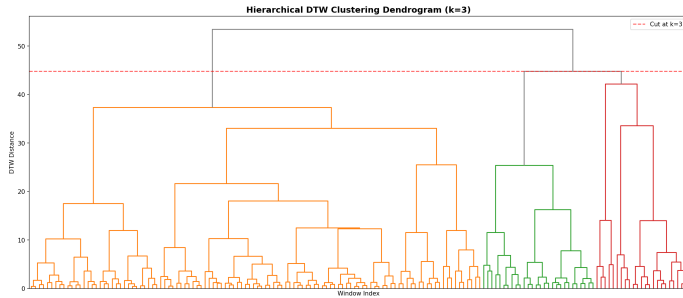


Figure 6. Ward-linkage DTW hierarchical clustering dendrogram ($k = 3$). The dashed horizontal cut identifies three structurally distinct regimes in the 21-day rolling window space.

Figure 7 overlays the DTW cluster assignments on the SPY price series. Clusters display multi-week temporal persistence, a clear improvement over the daily-resolution noise observed in static K-Means.

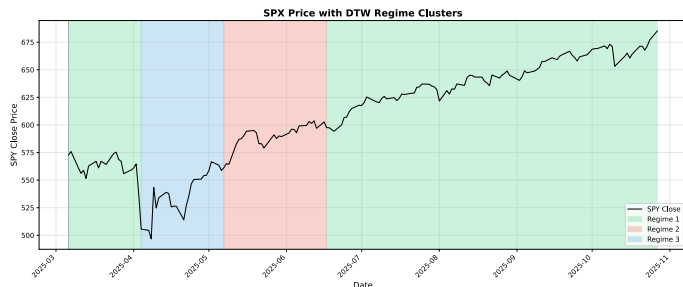


Figure 7. SPY price evolution with DTW cluster assignments. Multi-day temporal persistence is visible across all three clusters.

9.3. HMM: Probabilistic Persistence via the Transition Matrix

While DTW clustering provides a natural improvement over static methods, the HMM offers a more principled framework for two reasons. First, regime persistence is modeled explicitly through the transition probability matrix $\hat{\mathbf{A}}$, which is estimated jointly with emission

parameters from the full likelihood—rather than imposed implicitly through window length choice. Second, the HMM provides a well-defined posterior probability $P(S_t = j \mid \mathbf{y}_{1:T})$ for each observation, enabling soft regime assignment and uncertainty quantification that DTW clustering cannot deliver.

9.4. Comparative Summary

Table 7 summarizes the methodological trade-offs between the two approaches.

Table 7. HMM vs. DTW Hierarchical Clustering: Methodological Comparison

Property	HMM	DTW Clustering
Temporal persistence	Explicit ($\hat{\mathbf{A}}$)	Implicit (window)
Probabilistic output	Yes (posterior P)	No
Real-time applicability	Yes (filtering)	Limited
Look-ahead bias risk	Low	Low
Sensitivity to init.	Moderate	Low
Interpretability	High	Moderate
Computational cost	Moderate	High (DTW matrix)

10. DISCUSSION

10.1. Economic Validity

The three-state HMM recovers regime assignments that align with known economic events in the sample period. The Bull regime corresponds to periods of sustained SPY appreciation, low VIX, and a positively-sloped yield curve. The Bear regime clusters around periods of elevated VIX, negative return momentum, widening BAA–AAA spreads, and, in some instances, yield curve inversion. The Neutral regime captures transitional periods, consistent with the interpretation of markets drifting between structural states.

10.2. Practical Implications for Intraday Momentum

From a strategy perspective, the regime detector is intended to condition the intraday momentum signal. In the Bull regime, momentum signals may be acted on with full position sizing. In the Neutral regime, a reduced position cap is recommended. In the Bear regime, the strategy is gated off entirely or reversed, given the empirical breakdown of momentum in high-volatility regimes documented in the literature [Wang2020].

10.3. Limitations

Several limitations warrant acknowledgment. First, the Gaussian emission assumption underestimates fat-tailed return dynamics, particularly in the Bear regime where daily return distributions are more leptokurtic. A Student- t or variance-gamma emission would provide a more faithful distributional fit at the cost of additional model complexity. Second, the time-homogeneous transition matrix $\mathbf{A}$ cannot capture structural breaks or slow-moving parameter drift across economic cycles. Rolling-window or expanding-window re-estimation is advisable in live deployment. Third, the sample period of approximately one year following data intersection is relatively short, limiting the number of full bear-to-bull regime cycles observed; additional data would strengthen the statistical conclusions. Fourth, the DTW distance matrix computation scales as $O(N_w^2 \cdot W^2)$ where W is the window length, rendering it computationally prohibitive for very long samples without approximation.

10.4. Extensions

Natural extensions include: (i) Student- t or regime-switching GARCH emissions for heavier tails; (ii) time-varying transition matrices driven by macro state variables; (iii) online regime filtering using the forward

algorithm for real-time deployment; and (iv) integration of higher-frequency microstructure features (bid-ask spread, order imbalance) to improve intraday regime resolution.

11. CONCLUSION

We have presented a complete, reproducible market regime detection pipeline for the S&P 500 ETF applied to approximately 800 trading days of multi-source financial and macroeconomic data. Our methodology—comprising multi-source data acquisition, category-stratified PCA orthogonalization, multi-start Gaussian HMM training with penalized BIC order selection, and a DTW hierarchical clustering benchmark—delivers robust, economically interpretable regime labels suitable for downstream adaptive trading strategy components.

The optimal three-state HMM identifies persistent Bull, Neutral, and Bear regimes consistent with known market stress episodes in the sample. Regime-conditional statistics confirm that the model captures genuine structural volatility heterogeneity: the Bear regime exhibits annualized realized volatility approximately 2.3× the Bull baseline, with negative mean returns and widening credit spreads. The penalized BIC criterion successfully prevents economically degenerate short-lived states that standard BIC tends to select in financial time series.

The comparative analysis demonstrates that DTW hierarchical clustering, operating on 21-day rolling windows, improves materially over static K-Means by introducing temporal smoothing. However, the Gaussian HMM remains the superior framework for this application, offering explicit probabilistic persistence via the transition matrix, well-calibrated posterior regime probabilities for uncertainty quantification, and a principled real-time filtering procedure via the forward algorithm.

These findings validate the HMM pipeline as a suitable conditioning layer for the broader intraday momentum strategy, with clear operational protocols: full sizing in Bull regimes, reduced exposure in Neutral regimes, and strategy gating in Bear regimes.

A. REPOSITORY STRUCTURE AND EXECUTION

A.1. Folder Structure

The pipeline is executed via `uv run <script>`. The repository is organized as follows:

```
/data
  /raw/
  /cleaned/
  /processed/
  /reports/figures/
/src
  /classes/
    data/data_manager.py
    metrics/
      preprocessing.py
      curve_credit.py
      verify_financial_hypotheses.py
    clustering/regime_clustering.py
  /model/
    hmm_package/
      model.py
      main.py
      vis.py
```

Code 1. Repository folder structure.

A.2. Key Dependencies

Table 8. Core Python Dependencies

Package	Purpose
hmmlearn	Gaussian HMM estimation (Baum–Welch, Viterbi)
dtaidistance	Dynamic Time Warping distance computation
sklearn	PCA, preprocessing, silhouette scoring
pandas-datareader	FRED API data retrieval
yfinance	Yahoo Finance market data retrieval
statsmodels	ADF and KPSS stationarity tests